Sahaj Bhadja

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Summary

Android engineer pursuing an MSCS at USC with 2+ years building high-performance native apps in **Kotlin** and **Jetpack Compose**. Experienced in complex note-taking, annotation, and learning workflows with strong CS fundamentals, pragmatic engineering, and collaborative code reviews.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Operating Systems, Software Engineering, Computer Networks

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Structures

Experience

USC Dornsife **Los Angeles, CA**
Android Engineer *September 2025 – Present*

- Developed a native **Kotlin** tablet client with **Jetpack Compose** for an AI education platform, implementing immutable UI state, coroutine-driven sync, and performant note organization flows used by **500+ students** across large document libraries.
- Optimized rendering and memory usage for handwriting preview surfaces and PDF annotation views, reducing scroll jank and improving frame stability while participating in code reviews and communicating technical tradeoffs to cross-functional teammates.

Grad8 **Los Angeles, CA**
Android Engineer Intern *June 2025 – August 2025*

- Shipped **Kotlin** instructor features with **Jetpack Compose** for handwritten submission capture, rubric review, and grading history, integrating camera and stylus input paths that processed **10K+ daily requests** with responsive offline-first caching.
- Architected MVVM modules with structured concurrency and unit-tested repositories, collaborating in code reviews on Android best practices while debugging performance bottlenecks across large assessment files and peak classroom usage.

Aasma Technology Solutions **India**
Android Engineer - (SecureSync) *August 2023 – July 2024*

- Built and maintained a native **Kotlin** Android sync client for a cross-platform data-sharing product, implementing REST integrations, background workers, and resilient file transfer flows optimized for low-bandwidth users across production releases.
- Improved app startup and list rendering performance by **30%** through pagination, diffing, and lazy loading patterns while shipping well-tested releases via Gradle CI pipelines and Agile code review practices on a large Android codebase.

Aasma Technology Solutions **India**
Android Engineer Intern - (InkPad) *May 2023 – August 2023*

- Implemented **Jetpack Compose** note canvas modules with stylus stroke capture, layer management, and export workflows, applying functional UI patterns and immutable state models for a learning-focused Android prototype used in pilot classrooms.
- Integrated Room persistence and Firebase sync for notebook metadata, writing Espresso-style UI tests and resolving difficult threading issues across autosave, search indexing, and offline editing scenarios in a fast-paced product team.

LST Capitals **India**
Android Engineer Intern *November 2022 – January 2023*

- Developed high-performance **Kotlin** Android booking flows with Material Design components, real-time Firebase status updates, and wallet authentication, delivering sub-**200ms** interaction latency across **50K+ booking records** on production devices.
- Participated in architecture discussions and code reviews for a complex mobile codebase, debugging lifecycle edge cases and shipping pragmatic solutions that improved cross-device compatibility by **15%** under tight release deadlines.

Skills

Android: Kotlin, Jetpack Compose, Android SDK, Android Studio, Material Design, MVVM, Coroutines, Flow, Room, Gradle, RecyclerView (familiar), performance optimization, memory management, large codebases
Engineering: immutable data, functional patterns, REST APIs, Firebase, Git, CI/CD, unit testing, Espresso (familiar), Java, problem solving, code reviews, collaboration, Agile, pragmatic engineering, written communication

Projects

InkPad Code | *Kotlin, Jetpack Compose, Stylus, Room*

- Built a native note-taking Android app with **Jetpack Compose** canvas rendering, stylus stroke smoothing, folder organization, and PDF import/export, using immutable state and coroutines for responsive editing across large notebooks.

AnnotateFlow Code | *Kotlin, Compose, Handwriting, Learning UX*

- Designed an annotation-first Android experience for students with layered ink, searchable handwriting metadata, and low-latency scrolling optimizations, emphasizing empathy-driven UX patterns for organizing ideas and deepening understanding.

PiXMod Live | *Image Processing, Mobile-Ready APIs, Performance*

- Shipped image segmentation and editing services with performant API integrations and caching strategies, informing mobile-friendly asset pipelines and optimization techniques applicable to media-heavy Android learning and collaboration features.